



DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
BANGLADESH UNIVERSITY OF BUSINESS AND TECHNOLOGY

COURSE OUTLINE

1	Faculty	Faculty of Engineering & Applied Sciences (FEAS)
2	Department	Department of Computer Science and Engineering (CSE)
3	Program	Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE)
4	Course Code	PHY 101 (BNQF Code:05331101)
5	Course Title	Physics I
6	Course Type	Basic Science Course
7	Pre-requisites	N/A
8	Credit Hours/Value	3.0
9	Contact Hours	17 weeks × 2 classes per week × 1.25 hours per class = 42.5 total hours
10	Year-Semester	1-1
11	Academic Session	Spring 2024

12	Class Schedule	Intake – Section (Shift)	Class Day	Class Hours	Venue
		53-1(Day)	Sunday	9:15 AM-10:30 AM	Room: 906 (B-2)
			Thursday	10:30 AM-11:45 PM	Room: 910(B-2)
13	Course Website	Google Classroom Code: Meet Link: https://meet.google.com/rcy-dobm-fdn			

14	Course Teacher's Information	Name (Code): Md. Sharif Ahmed	Specialization: Material Science, Solid State physics, condensed materials, Nuclear Physics, etc.		
		Designation: Lecturer	Email: sharifahmed@gmail.com		
			Cell No. 01567828688	Room No. B1/602	

15	Counselling Hour/Tutorial	Day	Counseling Hours	Venue
		Sunday	1:30 pm -3:00 Pm	B1/602

16	Accessories & Aids	Students must carry learning materials like lecture notes, calculator, pen, pencil, eraser etc. in the classroom. Borrowing learning materials in the classroom or exam room from fellow students is prohibited. A student is also advised to keep a separate class note (khata) of 50 pages for the course during class hours.
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17	Course Rationale	This course will give an overview and understanding of basic Physics with moderate use of mathematical formalism and a quick introduction to waves and oscillations, thermodynamics, optics and nature of light.
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18	Course Objective	The Course has therefore been designed to fulfill the following objectives: Obj.1. To teach students the fundamental theory of Simple harmonic motion, combination of simple harmonic oscillation, and Damped oscillation. Obj.2. To apply heat and thermodynamics theories for practical problem solving's. Obj.3. To introduce the students with advance features of fundamentals theory of Physics.
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19	Course Learning Outcomes (CLOs)	Upon completing this course students will be able to: CO1: Characterize the concepts of waves and oscillations, thermodynamics' laws, crystal, interference, diffraction and polarization, etc.; CO2: Explain physical natural phenomena related to waves and oscillations, thermodynamics, interference, diffraction and polarization in a clear manner; CO3: Solve problems in the areas of waves and oscillations, thermodynamics, interference, diffraction and polarization, etc.
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20	Mapping of CO – PO – K, P/A – Learning Domain, and Method of Delivery and Assessment Tool					
	CLOs	PLOs & CF	Learning Domain	Ks & Ps**	Delivery Methods / Activities	Assessment Tools
	CLO1	PLO1 CF=2	Cognitive / Understanding	K1, K2 & K3	- Classroom Lecture (PPP &/or WBT*) - Classroom Discussion & Exercise practice - Analysis and design problem solving - Course Counseling	Indirect: In class response, course counseling, course end survey Direct: Class participation and activity, class test, assignment, midterm and final examinations
	CLO2	PLO1 CF=2	Cognitive / Analyzing	K1, K2 & K3; P1		
	CLO3	PLO1 CF=2	Cognitive / applying	K1, K2 & K3; P1		
		*PPP: Power Point Presentation & WBT: White Board Teaching **Ks: Knowledge Profiles & Ps: Profiles of Complex Engineering Problem				
	Correlation of COs to POs					
Correlation Criteria				Correlation Level	Correlation Factor (CF)	
Less than 25%				Almost no correlation	0	
Equal to or greater than 25% and below 50%				Poor (Low)	1	
Equal to or greater than 50% and below 75%				Moderate (Medium)	2	
Equal to or greater than 75% and up to 100%				Significant (High)	3	

21	Teaching-Learning Strategy	Maximum topics will be covered from the textbook. For the rest of the topics, reference books will be followed. Some class notes will be uploaded on the web. White board will be used for most of the time. For some cases, multimedia projector will be used for the convenience of the students. Students must participate in classroom discussions for case studies, problems solving and project developments.
22	Course Plan	This course consists of 150.0 min./week of class contact hours and an additional 90 min./week of counseling hours to explain students' design problems, provide reading materials, and assist in understanding lecture materials for preparing examinations, class tests, and assignments.

Week No.	Lecture No.	Lecture Topic (syllabus covered)	Lecture Note and Text Book	Teaching-Learning Strategy	Assessment Strategy	CO
1	1	Setting Context of Teaching and Learning Based on OBE Model Overview of communication systems: Waves and Oscillations: periodic motion, oscillatory motion, simple harmonic motion, characteristics of SHM.	Course profile Hand notes and Chapter 1[Text 1] Chapter 1: Prob.: 1.1,1.2, 1.3, 1.4, 1.5	Delivering presentation-based instruction Delivering lecture-based instruction, classroom discussion and example problem solving	Class test 1 & Midterm Exam.	CO1, CO2, CO3
	2					
2	3	Differential Equation of a simple harmonic oscillator and solution of it, Total energy and average energy, combination of simple harmonic oscillation.	Hand notes and Chapter 1 [Text 1] Chapter 1: Prob.: 1.6, 1.7, 1.8, Class Test 01	Delivering lecture-based instruction, classroom discussion, example problem solving and assign assignment and disclose in a presentation	Assignment & Midterm Exam.	CO1, CO2, CO3
	4					
3	5	Damped oscillation, undamped oscillation, Resonance, problems, Heat & Thermodynamics: Heat, Temperature. Different types of thermometer, Gas laws, Equation of state for a perfect gas, Absolute zero temperature, Standard Temperature and pressure.	Hand notes Class test 1 Chapter 1: Prob.: 1.9, 1.10	Delivering lecture-based instruction, classroom discussion, example problem solving and assign assignment and disclose in a presentation	Class test 1 & Midterm Exam.	CO1, CO2, CO3
	6					
4	7	Damped oscillation, undamped oscillation, Resonance, problems, Heat & Thermodynamics: Heat, Temperature.	Hand notes Text 1: Chapter 19, Prob.: 16.1, 16.2, 16.3, 16.4, 16.8, 16.11, 16.14, 16.15,16.16	Delivering lecture-based instruction, classroom discussion, example problem solving and assign assignment and disclose in a presentation		
	8					
5	9	Kinetic theory of gases, Mean free path, mathematical expression of Mean free path, mean free path, Equipartition of energy, Brownian motion, Van der Waal's equation of state. Zeroth law and 1st law of thermodynamics, Applications of 1st law of thermodynamics, 2nd laws of thermodynamics and mathematical problems.	Hand Notes Text 1: Chapter 19, Prob.: 15.17, 15.18, 15.19, 15.20, 15.21	Delivering lecture-based instruction, classroom discussion and example problem solving	Class test 1 & Midterm Exam.	
	10					

6	11	Kinetic theory of gases, Mean free path, mathematical expression of Mean free path, mean free path, Equipartition of energy, Brownian motion, Van der Waal’s equation of state.	Hand Notes Class Test 02 Text 1: Chapter 20, Prob.: 20.1-20.3	Delivering lecture-based instruction, classroom discussion, example problem solving and assignment and disclose in a presentation	Assignment & Midterm Exam.	CO1, CO2, CO3
	12					
7	13	Heat engine, Carnot engine, Efficiency of heat engine, mathematical problem of Carnot engine and 1 st law of thermodynamics, application of thermodynamics. Entropy, Change in entropy in adiabatic and isothermal process, application of entropy, entropy and Third Law of thermodynamics and mathematical problems.	Hand Note Text 1: Chapter 20, Prob.: 20.3-20.8	Classroom discussion and problem solving	Midterm Exam.	CO1, CO2, CO3
	14					
8	15	Quick Recap/Summary of whole lectures of Midterm	Hand note. Text 1: Chapter 20, Prob.: 20.9-20.13	Delivering presentation-based instruction	Midterm Exam.	CO1, CO2, CO3
	16					
Midterm Examination (according to the Academic Calendar- and examination schedule)						
9	17	Structure of Matter: Crystalline & non-crystalline solids, single crystal and poly crystal solids, lattice, basis and basis vectors.	Hand Notes Chapter 19 [Text 1] Text 1: Chapter 19, Prob.: 19.1, 19.2, 19.3, 19.4, 19.5,19.6,19.7	Delivering lecture-based instruction, classroom discussion, example problem solving and assignment and disclose in a presentation	Assignment Final Exam	CO1, CO2, CO3
	18					
10	19	Unit cell, crystal systems in two and three dimensions., Miller indices, Packing factor, Bragg’s law, Bond in solid and bond energy. Band theory of solid, Defects in solids: point defects, line defects; Bonds in solids, inter-atomic distances, calculation of cohesive.	Hand Notes Chapter 19 [Text 1] Text 1: Chapter 19, Prob.: 19.8,19.9, 19.10, 19.11, 19.12	Delivering lecture-based instruction, classroom discussion, example problem solving and assignment and disclose in a presentation		
	20					
11	21	Bounding energy; Introduction to band theory: distinction between metal, semiconductor and insulator. Unit cell, crystal systems in two and three dimensions., Miller indices,	Hand Notes Chapter 19 [Text 1] Text 1: Chapter 19, Prob.:	Delivering lecture-based instruction, classroom discussion and	Class Test 2 Final Exam	
	22					

			Packing factor, Bragg's law, Bond in solid and bond energy.	19.12, 19.13, 19.14,19.15,	example problem Delivering lecture-based instruction, classroom discussion, example problem solving and assign assignment and disclose in a presentation		
12	23		Band theory of solid, Defects in solids: point defects, line defects; Bonds in solids, inter-atomic distances, calculation of cohesive.	Hand Notes Chapter 19 [Text 1] Text 1: Chapter 19, Prob.: 19.16, 19.17, 19.18, 19.19, 19.20			
	24						
13	25		Interference: Coherent sources, Young's double slit Experiment, Analysis of interference, Fringes, problems. Fresnel's biprism, Newton's ring, interferometers, problems.	Hand Notes Chapter 27 [Text 1] Text 1: Chapter 27, Prob.: 27.1, 27.2, 27.3, 27.4, 27.5 Class Test 03	Delivering lecture-based instruction, classroom discussion, example problem solving and assign assignment and disclose in a presentation		
	26						
14	27		Diffraction: Diffraction by single slit, Position of maxima-minima, problems.	Hand Notes [Text 1] Chapter 27 Prob.: 27.16 to 27.27	Assignment submission with a presentation	Class Test 3	CLO2
	28						
15	29		Double slit Fraunhoffer diffraction, Diffraction grating, , Resolving power of grating, and Fresnel diffraction by circular aperture.	Hand Notes Text 1: Chap.28, Prob.: 28.1, 28.2, 28.3, 28.4, 28.5	Delivering lecture-based instruction, classroom discussion and example problem solving	Assignment 2	CLO2 & CLO3
	30						
16	31		Polarization: production and analysis of polarized light, Brewster's law, Malus' law, Polarization by double refraction, retardation plates, Nicol prism, optical activity, Polarimeters, problems.	Hand note. Text 1: Chap.30, Prob.: 30.1, 30.2, 30.3, 30.4 and 30.10, 30.11, 30.12	Delivering lecture-based instruction, classroom discussion, example problem solving and assign assignment and disclose in a presentation	Assignment Final Exam	CLO1 & CLO2
	32						
17	33		Quick Recap/Summary of whole lectures of Final.	--	Delivering presentation-based instruction	Final Exam	CO1, CO2, CO3
	34						
Final Examination (according to the Academic Calendar and examination schedule)							

23 i	Class Note	1. Class Note of the course of Physics I for the students of B.Sc. in Electrical and Electronic Engineering (EEE) Program, BUBT by Mohammad Jahidul Islam
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ii	Text Books	[1] Dr. Giasuddin Ahmad “Physics for Engineers” (part 1 & 2) [1] Fundamental of Physics”, Halliday, Resnick and Walker, 10 th Edition [2] "Physics”, Halliday, Resnick and Krane, Fifth Edition, Volume II [3] “Concepts of Modern Physics”, Arthur Beiser [4] “Modern Physics”, Kenneth S. Krane, 3 rd Edittion.																																															
24	Assessment and marks distribution criteria	An integral part of this course is engagement in class activities, outside classroom discussions and communications through net and phone. Students may fail the course if they do not participate in the class regularly, class tests and/or complete assignments. Students must actively participate in classroom discussions and complete all sets of works at a satisfactory level, as mentioned in this course outline. The course-specific expectations for students are achieved if 75% of students in a section attend more than 70% of classes (determined by summative assessment). - Their active participation in the classroom discussion is targeted up to 80% of total attendee (determined by formative assessment such as eye-contact, clicker questions and group discussions). - Equal or more than 40% of course outcomes must be achieved by the students (summative assessment). - The level of engagement in the studies such as preparing regularly class lectures, class tests, assignments must be more than 60% (formative assessment). - Expected level of participation in the outside class discussion (once in a weak, more than 30% of students in a section) by course counseling and using social media like Google Classroom, email and phone call etc. (formative assessment). - Students are assessed according to their individual performance in the examinations, class tests, assignments, and class participations. The final mark calculation and course outcome assessment are done based on the following mark distribution (criteria): <table><tr><th>Assessment tool</th><th>Conducting Number</th><th>Mark distribution (%)</th></tr><tr><td>Class participation and activity</td><td>34</td><td>05</td></tr><tr><td>Class test</td><td>2</td><td>7.5 × 2 = 15</td></tr><tr><td>Assignments/Report and Presentation</td><td>1</td><td>5.0 + 5.0 = 10</td></tr><tr><td>Midterm examination</td><td>1</td><td>30</td></tr><tr><td>Final examination</td><td>1</td><td>40</td></tr><tr><td colspan="2">Total Mark</td><td>100</td></tr><tr><th colspan="3">Class Participation & Activity Performance Criteria</th></tr><tr><th colspan="2">Performance level</th><th>Mark distribution (%)</th></tr><tr><td colspan="2">90% -100%</td><td>05</td></tr><tr><td colspan="2">80% - 89%</td><td>04</td></tr><tr><td colspan="2">70% - 79%</td><td>03</td></tr><tr><td colspan="2">60% - 69%</td><td>02</td></tr><tr><td colspan="2">50% - 59%</td><td>01</td></tr><tr><td colspan="2">Less than 50%</td><td>Not allowed to sit for the final examination.</td></tr></table>			Assessment tool	Conducting Number	Mark distribution (%)	Class participation and activity	34	05	Class test	2	7.5 × 2 = 15	Assignments/Report and Presentation	1	5.0 + 5.0 = 10	Midterm examination	1	30	Final examination	1	40	Total Mark		100	Class Participation & Activity Performance Criteria			Performance level		Mark distribution (%)	90% -100%		05	80% - 89%		04	70% - 79%		03	60% - 69%		02	50% - 59%		01	Less than 50%		Not allowed to sit for the final examination.
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25	Assessment Strategy																																																
	Assessment tool	Content, Length and Criteria	Weight (mark)	Due date																																													

Class test 1	Composed of 4 to 6 short questions which cover the topic of waves and oscillation, kinetic energy of oscillating particle. Duration of the class test is 40 minutes, and the assigned full mark is 7.5.	7.5	On the 12th lecture
Class test 2	Composed of 4 to 6 short questions which cover the topics of heat and thermodynamics. Duration of the class test is 40 minutes, and the assigned full mark is 7.5.	7.5	On the 25th lecture
Assignment with a presentation	Kinetic theory of gasses, heat engine and efficiency of heat engine and their applications. Assignment with a presentation will be delivered two weeks after assigning. 5 marks are assigned for the assignment and the other 5 marks are for a presentation.	10.0	Presentation will be conducted on the 28th lecture
Midterm Examination	Midterm Examination is held according to the Academic Calendar and examination schedule after 8 weeks of classes. All three sets of questions are to be answered by the students in one and half-hour exam period. Each question set contains multiple short type questions. The assigned full mark for the 3 sets of questions is 30.	30	After 16 th lecture and following the examination schedule of BUBT.
Final Examination	Final Examination is held after 8 weeks of classes after the midterm examination. All four sets of questions are to be answered by the students in a two-hour exam period. Each question set contains multiple short type questions. The assigned full mark for the 4 sets of question is 40.	40	After 9 weeks of classes afterward the midterm examination and following the examination schedule of BUBT.

CLO Assessment Criteria	Assessment of CLOs					
	Assessment Tool	CLO				Mark Allocation
		CLO1	CLO2	CLO3		
	Class Participation	5/3	5/3	5/3		05
	Class Test	4.0	6.0	5.0		15
	Assignment	0	5.0	5.0		10
	Midterm Exam	10	20	0		30
	Final Exam	05	12	13		40

27	Rubrics (Attainment Criteria)	CLOs (Taxonomy domain)	Not attained /Failed (0-39%)	Poor (40%-49%)	Moderate (50%-64%)	Good (65%-79%)	Excellent (80%-100%)
		CLO1 (understand) CLO2 (understand and apply) CLO3 (apply) CLO4 (understand and apply)	The question is answered with serious deficiencies in understanding and explanation. Applicable method is not almost touched.	The question is answered inadequately by touching the applicable method or without explanations. As a result, a few steps of problem-solving procedure or concepts are not developed properly or missing.	The question is answered partially correct by applying the asking method or concepts, but a few important details are missing.	The question is answered correctly but briefly missing some portions of the important explanation by applying the required method or concepts.	The question is answered correctly with detail explanations using the asking method of solving the problem or concepts with adequate explanation.
28	Feedback	All kinds of feedback to the students will be produced within a week after the day of holding a class test and midterm examination. No answer script will be shown for the final examination if not challenged by a student. Online/email queries will also be responded within three days by email.					
29	Grading Policy	Letter grades and grade points are used to evaluate the performance of a student in the course:					
		Marks Range		Letter Grade		Grade Point	
		80% and above		A+	: A Plus (Outstanding)	4.00	
		75% to less than 80%		A	: A regular (Excellent)	3.75	
		70% to less than 75%		A-	: A minus (Very good)	3.50	
		65% to less than 70%		B+	: B Plus (Good)	3.25	
		60% to less than 65%		B	: B regular (Average)	3.00	
		55% to less than 60%		B-	: B minus (Below average)	2.75	
		50% to less than 55%		C+	: C Plus (Poor)	2.50	
		45% to less than 50%		C	: C regular (Very poor)	2.25	
		40% to less than 45%		D	: Pass marginally	2.00	
		Less than 40%		F	: Fail	0.00	
				I	: Incomplete		
				R	: Retake		
				W	: Withdraw		
30	Additional Course Policies	Assignment	All assignments have equal weights. No late submission will be accepted. Any kind of copied information without proper citation (i.e., plagiarism) in the assignments or any other work is prohibited and treating as a serious academic misconduct which is prosecuted through the Discipline Committee of BUBT. All copied/plagiarized assignments/reports/test papers will be canceled, and the isolated student must prepare a new assignment/project again. Zero tolerance will be shown in this regard. Feedback after submission of long assignments will be provided on hand or by email within two weeks.				
		Class Test	There will be conducted two class tests (CTs) with one quiz or three CTs. All class tests/quizzes have equal weightage. Both regular and surprise CTs/quizzes can be conducted.				
		Closed book assessments	Quizzes, CTs, midterm and final examinations are closed book assessments. Mobile phone is prohibited in the examination hall. Students are insisted on carrying simple scientific calculators to solve the complex calculations and a wrist-watch to follow time during the exam hours.				
		Test Policy	If a student is absent from class test anyway and has made no report to the class teacher personally beforehand, his/her score for that test will be zero. No make-up for the class test will be allowed. No supplementary for midterm and final examinations will be entertained without physical presence and recommendation of the guardian, along with written permission of the department. Supplementary examination questions are much				

			harder than the regular examination questions; therefore, students are discouraged from taking supplementary examinations.
31	Additional Information	Academic Calendar Fall 2023: https://www.bubt.edu.bd/Home/page_details/Academic_Calender Academic Rules: https://www.bubt.edu.bd/Home/page_details/Rules_and_Regulations Grading & Evaluation: https://www.bubt.edu.bd/Home/page_details/Evaluation_Grading_System Rules & Regulations: https://www.bubt.edu.bd/Home/page_details/Office_of_the_Registrar	
32	Bloom’s Taxonomy for Teaching-Learning Bloom's Taxonomy is a set of three hierarchical models used to classify educational learning objectives into levels of complexity and specific quality. The three taxonomy domains for achieving learning objectives are cognitive, affective, and psychomotor. Cognitive domain is in the primary focus of educating and frequently used to structure curriculum learning objectives and achieve the level of learning. The three domains and their respective levels are illustrated below:		
	Cognitive [C] (Knowledge-based)	Affective [A] (Emotion-based)	Psychomotor [P] (Action-based)
	Remember	Receive	Imitate
	Understand	Respond	Manipulate
	Apply	Value	Precision
	Analyze	Organize	Articulation
	Evaluate	Characterize	Naturalization
	Create	--- --- ---	--- --- ---
33	Descriptions of Cognitive Domain (Anderson and Krathwohl’s updated Taxonomy in 2001): The cognitive domain involves the development of our mental skills and the acquisition of knowledge.		
	Category (Level)	Meaning	Keywords
	Remember (C1)	Recognizing or recalling knowledge from memory. Remembering is when memory is used to produce or retrieve definitions, facts, or lists, or to recite previously learned information.	Define, describe, draw, find, identify, label, list, match, name, quote, recall, recite, tell, and write
	Understand (C2)	Constructing meaning from different types of functions be they written or graphic messages or activities like interpreting, exemplifying, classifying, summarizing, inferring, comparing, or explaining.	Classify, compare, exemplify, conclude, demonstrate, discuss, explain, identify, illustrate, interpret, paraphrase, predict, and report
	Apply (C3)	Carrying out or using a procedure through executing or implementing. Applying relates to or refers to situations where learned material is used through products like models, presentations, interviews, or simulations.	Apply, change, choose, compute, dramatize, implement, interview, prepare, produce, role play, select, show, transfer, and use
	Analyze (C4)	Breaking materials or concepts into parts, determining how the parts relate to one another or how they interrelate, or how the parts relate to an overall structure or purpose. Mental actions included in this function are differentiating, organizing, and attributing, as well as being able to distinguish between the components or parts. When one is analyzing, he/she can illustrate this mental function by creating spreadsheets, surveys, charts, or diagrams, or graphic representations.	Analyze, characterize, classify, compare, contrast, debate, deconstruct, deduce, differentiate, discriminate, distinguish, examine, organize, outline, relate, research, separate, and structure
	Evaluate (C5)	Making judgments based on criteria and standards through checking and critiquing. Critiques, recommendations, and reports are some of the products that can be created to demonstrate the processes of evaluation.	Appraise, argue, assess, choose, conclude, decide, evaluate, judge, justify, predict, prioritize, prove, rank, rate, select, Monitor.

	Create (C6)	Putting elements together to form a coherent or functional whole; reorganizing elements into a new pattern or structure through generating, planning, or producing. Creating requires users to put parts together in a new way, or synthesize parts into something new and different creating a new form or product. This process is the most difficult mental function in the new taxonomy.	Create, invent, compose, predict, plan, construct, design, propose, devise, and formulate
34	Graduate Attributes (Program Learning Outcomes) of B.Sc. in EEE Program based on Washington Accord		
	Program Learning Outcomes (PLOs) are brief statements that describe what students are expected to know and be able to do by the time of graduation. These relate to the knowledge, skills, and attitudes that students acquire throughout the entire course of a program. The students of the B.Sc. in CE program are expected to achieve the following graduate attributes or program outcomes at the time of graduation: PLO1: (Engineering knowledge): Apply knowledge of mathematics, natural science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems. PLO2: (Problem analysis): Identify, formulate, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics and natural and engineering sciences. PLO3: (Design/development of solutions): Design solutions for complex engineering problems and design systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations. PLO4: (Investigation): Conduct investigations of complex problems using research-based knowledge and research methods that include the design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions. PLO5: (Modern tool usage): Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems, with an understanding of the limitations. PLO6: (The engineer and society): Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. PLO7: (Environment and sustainability): Understand and evaluate the sustainability and impact of professional engineering work to solve complex engineering problems in societal and environmental contexts. PLO8: (Ethics): Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice. PLO9: (Individual work and teamwork): Function effectively as individuals and members or leaders of diverse teams and in multidisciplinary settings. PLO10: (Communication): Communicate effectively on complex engineering activities with the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and convey and receive clear instructions. PLO11: (Project management and finance): Demonstrate knowledge and understanding of engineering and management principles and apply them to their work as team members or leaders or entrepreneurs to manage projects in multidisciplinary environments. PLO12: (Life-long learning): Recognize self-awareness to engage in a lifelong learning process to reflect in the broadest context of technological change.		
35	Knowledge Profile (K)		
	K1: A systematic, theory-based understanding of the natural sciences applicable to the discipline. K2: Conceptually-based mathematics, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modeling applicable to the discipline. K3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline. K4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline much is at the forefront of the discipline. K5: Knowledge that supports engineering design in a practice area. K6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.		

	K7: Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the professional responsibility of an engineer to public safety; the impacts of engineering activity: economic, social, cultural, environmental and sustainability. K8: Engagement with selected knowledge in the research literature of the discipline.
36	Range of Complex Engineering Problem Solving (P) Complex Engineering Problems have characteristics P1 and several or all of P2 to P7: P1: Depth of knowledge required: Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3, K4, K5, K6 or K8, which allows a fundamentals-based, first principles analytical approach P2: Range of conflicting requirements: Involve wide-ranging or conflicting technical, engineering and other issues P3: Depth of analysis required: Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models P4: Familiarity of issues: Involve infrequently encountered issues P5: Extent of applicable codes: Are outside problems encompassed by standards and codes of practice for professional engineering P6: Extent of stakeholder involvement and conflicting requirements: Involve diverse groups of stakeholders with widely varying needs P7: Interdependence: Are high-level problems including many component parts or sub-problems
37	Range of Complex Engineering Activities (EA) Attribute Complex activities means (engineering) activities or projects that have several or all of the following characteristics: A1: Range of resources: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies). A2: Level of interaction: Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering, or other issues. A3: Innovation: Involve creative use of engineering principles and research-based knowledge in novel ways. A4: Consequences for society and the environment: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation. A5: Familiarity: Can extend beyond previous experiences by applying principles-based approaches.
38	Code of Conduct It is strongly suggested that students keep discipline in the classroom attending on time, listening lecture attentively and participating in discussion on the subject. To get class participation grade, students MUST attend the classes of courses s/he registered for. Turn off his/her cell phone before enter in a class or participating in class tests and exams. There are activities which are considered as academic misconduct. One of them is plagiarism, which signifies the deliberate formal presentation or submission of works, phrases, texts, ideas, illustrations, or diagrams of others as one's own without proper citation. Another one is the use of unauthorized aids (including electronic devices), asking assistance or using illegal materials preparing assignments or in examinations. In addition, copying from other's work, showing your work to others and asking for answers are also considered as an academic misconduct. Penalties for involving academic misconduct include one or more of the following: a zero grade on the work produced, a failing grade in the course, suspension for one semester or more and even expulsion from the university. On the university premises or at a university-sponsored program, students must abide by the Student Code of Conduct and other Rules and Regulations of BUBT, which are available on the BUBT website at https://www.bubt.edu.bd/Home/page_details/Office_of_the_Proctor.
39	Social & Moral Values Our promises are based on the three cardinal principles: - What we do believe. - What we do practice. - What we will promote. However, students are advised to undertake the following commitments for social and moral developments.

	• To be punctual and attentive in classes; • To prioritize honesty & faith; • To ensure mutual respect; • To be always proactive; • To avoid conspiracy; • To be cooperative in learning; • To be sincere in class preparation;	• To avoid unfair means and plagiarism in exams, report writings and assignments; • To carry out assignments or keep other commitments timely; • To be motivated for asking question and encourage feedback; • Not to forget to switch-off the cellphone in a class;	• To follow the dress code and wearing ID card on campus; • To be decent on all aspects; • To be loyal and trust-worthy to the teachers and others; • Help keeping an eco-friendly environment in the campus.
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